

Session 13

1. Create a decision table for a Flipkart-style coupon validation system where a coupon is only valid if: (1) the user is logged in, (2) the cart value is above ₹500, and (3) the coupon has not expired. Show all possible combinations and the resulting action (accept/reject).

Decision Table – Flipkart-Style Coupon Validation

Conditions

A coupon is accepted only when **all three conditions are true**:

1. User is logged in.
2. Cart value is above ₹500.
3. Coupon has not expired.

Y = Yes/True, N = No/False

Conditions / Rules	R1	R2	R3	R4	R5	R6	R7	R8
User logged in?	Y	Y	Y	Y	N	N	N	N
Cart value > ₹500?	Y	Y	N	N	Y	Y	N	N
Coupon not expired?	Y	N	Y	N	Y	N	Y	N
Action	Accept	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Explanation

- **R1**: All conditions are satisfied → **Coupon Accepted**
- **R2**: Coupon expired → **Rejected**
- **R3**: Cart value is ₹500 or less → **Rejected**
- **R4**: Cart value is ₹500 or less and coupon expired → **Rejected**
- **R5–R8**: User is not logged in → **Rejected**

Example

If the user is logged in, cart value is ₹700, and coupon is active:

Logged in = Y + Cart > ₹500 = Y + Not expired = Y → **ACCEPT**

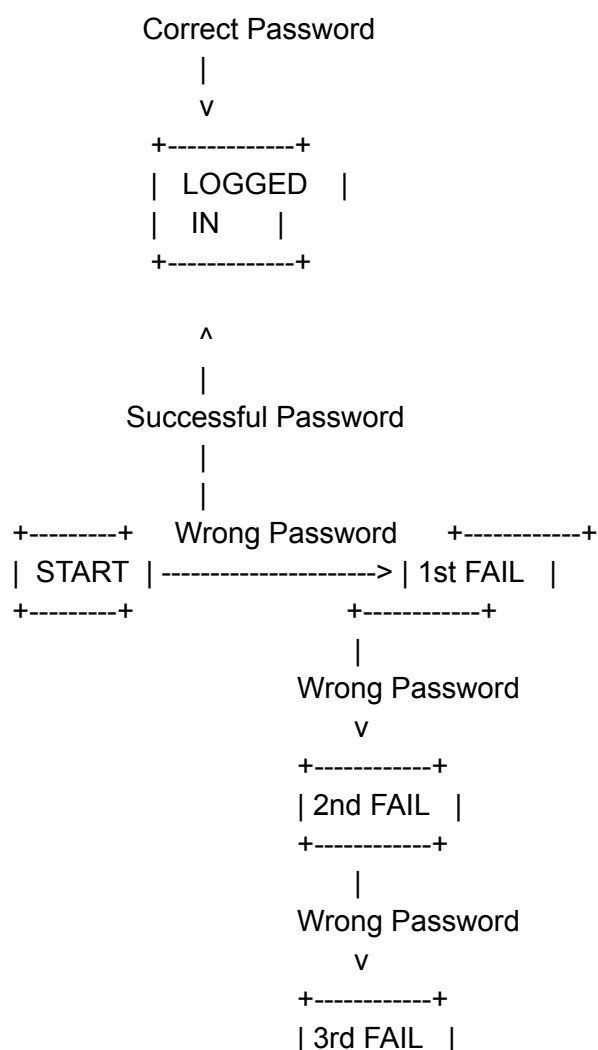
2. Draw a state transition diagram for a Paytm login feature that locks the account after 3 failed password attempts and only unlocks after a successful OTP verification.

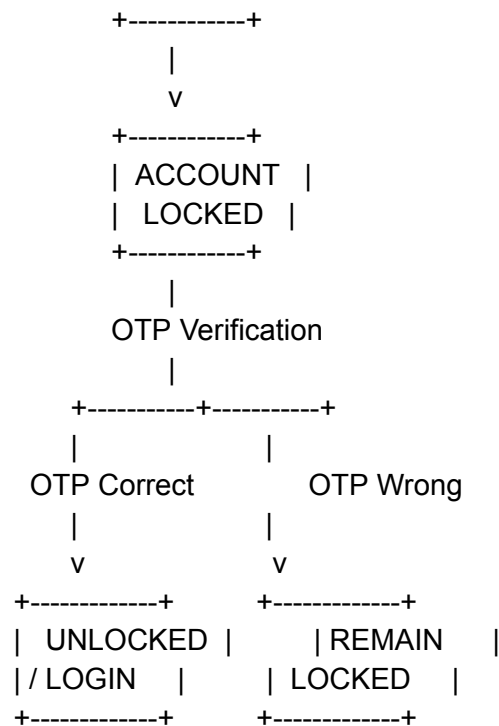
State Transition Diagram – Paytm Login

Requirement

- User enters password.
- If password is correct → login successful.
- If password is incorrect → failed attempt count increases.
- After **3 failed attempts**, account is locked.
- Locked account can only be unlocked after **successful OTP verification**.

State Diagram





Important Transitions

Current State	Event	Next State
Start	Correct password	Logged In
Start	Wrong password	1st Failed
1st Failed	Wrong password	2nd Failed
2nd Failed	Wrong password	3rd Failed
3rd Failed	Wrong password	Account Locked
Account Locked	Wrong OTP	Remains Locked
Account Locked	Correct OTP	Unlocked/Login

Test Scenarios

1. Enter the correct password → user should successfully log in.
2. Enter the wrong password three times → account should become locked.
3. Enter incorrect OTP after account lock → account should remain locked.
4. Enter correct OTP after account lock → account should be unlocked.

Given the following use case: 'A Zomato user can place an order, cancel it before it is prepared, or track the order once prepared,' create a use case diagram and list at least 3 possible test scenarios based on it.

Use Case Diagram – Zomato Order

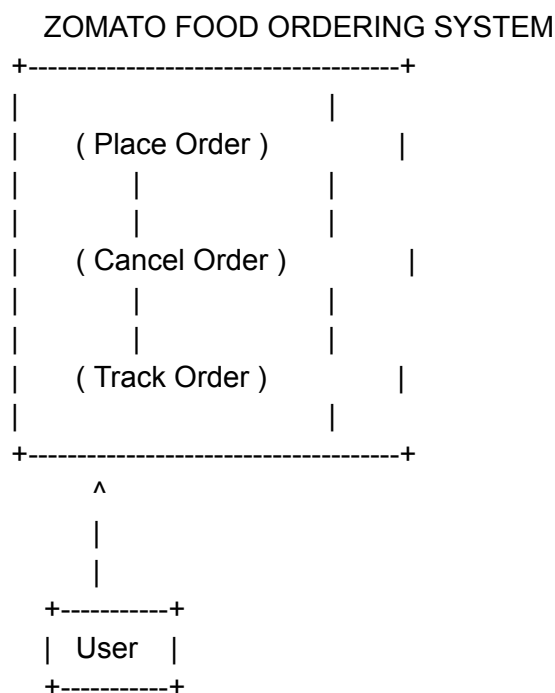
Actors and Use Cases

Actor: Zomato User

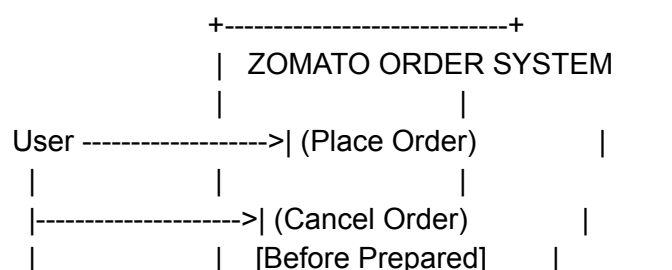
Use Cases:

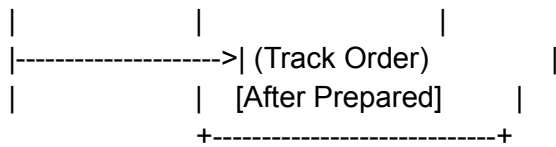
- Place Order
- Cancel Order
- Track Order

Use Case Diagram



More Detailed Relationship





Test Scenarios

Scenario 1 – Place Order

- User selects food.
- Adds food to cart.
- Makes payment.
- Clicks Place Order.
- **Expected:** Order should be successfully placed.

Scenario 2 – Cancel Order Before Preparation

- User places an order.
- Restaurant has not started preparing it.
- User selects Cancel Order.
- **Expected:** Order should be cancelled.

Scenario 3 – Track Prepared Order

- User places an order.
- Restaurant starts/prepares the order.
- User selects Track Order.
- **Expected:** Current order status/location should be displayed.

Scenario 4 – Cancel After Preparation

- Order is already being prepared.
- User tries to cancel.
- **Expected:** Cancellation should not be allowed if the requirement says cancellation is only before preparation.

4. Modify your decision table from Task 1 to include an additional rule: the coupon cannot be used more than twice by the same user. Update the table and explain how this new rule changes the possible outcomes.

Hint: Add a column for 'Coupon Usage Count' and adjust the actions accordingly.

Updated Decision Table – Coupon Usage Count

New Requirement

The coupon **cannot be used more than twice by the same user**.

So the coupon is accepted only when:

- User is logged in = **Yes**
- Cart value > ₹500 = **Yes**
- Coupon not expired = **Yes**
- Coupon usage count < 2 = **Yes**

Here, "usage count < 2" means the coupon has been used **0 or 1 times**. If the coupon has already been used **2 times**, another use is rejected.

Updated Decision Table

Conditions / Rules	R1	R2	R3	R4	R5
User logged in?	Y	Y	Y	Y	Y
Cart value > ₹500?	Y	Y	Y	Y	N
Coupon not expired?	Y	Y	N	Y	Y
Usage count < 2?	Y	N	Y	N	Y
Action	Accept	Reject	Reject	Reject	Reject

Explanation of the New Rule

Previously, a coupon could be accepted when the first three conditions were satisfied.

After adding the **usage count** condition:

All 4 conditions must be satisfied for acceptance.

For example:

Login	Cart	Expired?	Usage Count	Result
-------	------	----------	-------------	--------

Yes	₹700	No	0	Accept
Yes	₹700	No	1	Accept
Yes	₹700	No	2	Reject
Yes	₹700	No	3	Reject
No	₹700	No	0	Reject

Key Point

The new rule creates an **additional reason for rejection**. Even if the user is logged in, the cart is above ₹500, and the coupon is active, the coupon must be **rejected when its usage count has already reached 2**.

Final Conclusion

- **Decision Table:** Tests all combinations of conditions and determines Accept/Reject.
- **State Transition Testing:** Checks how the system moves between states based on events.
- **Use Case Testing:** Tests the functions available to the user from the use-case requirements.
- **EP/BVA:** Divide inputs into classes and focus on boundaries.
- These techniques help testers identify missing conditions, invalid transitions, and incorrect system behavior.